the fuel tank sender input to the PCM is an open circuit.

- **Communication Error** - If the cluster fails to receive a percent tank full message, it will hold the gauge needle at the last indication about twelve seconds or until the ignition switch is turned to the Off position, whichever occurs first. After twelve seconds, the cluster will move the gauge needle to the left end of the gauge scale and the low fuel indicator is illuminated immediately.
- **Actuator Test** - Each time the cluster is put through the actuator test, the gauge needle will be swept to several calibration points on the gauge scale in a prescribed sequence to confirm the functionality of the gauge and the cluster control circuitry.

The PCM continually monitors the fuel tank sending unit to determine the level of the fuel in the fuel tank. The PCM then sends the proper fuel level messages to the instrument cluster. For further diagnosis of the fuel gauge or the instrument cluster circuitry that controls the gauge, (Refer to 8 - ELECTRICAL/ INSTRUMENT CLUSTER - DIAGNOSIS AND TESTING). For proper diagnosis of the fuel tank sending unit, the PCM, the PCI data bus, or the electronic message inputs to the instrument cluster that control the fuel gauge, a DRBIII® scan tool is required. Refer to the appropriate diagnostic information.

HIGH BEAM INDICATOR

DESCRIPTION

Fig. 16 High Beam Indicator

A high beam indicator is standard equipment on all instrument clusters (Fig. 16). The high beam indicator is located near the upper edge of the instrument cluster, between the tachometer and the speedometer. The high beam indicator consists of a stencil-like cutout of the International Control and Display Symbol icon for “High Beam” in the opaque layer of the instrument cluster overlay. The dark outer layer of the overlay prevents the indicator from being clearly visible when it is not illuminated. A blue lens behind the cutout in the opaque layer of the overlay causes the icon to appear in blue through the translucent outer layer of the overlay when the indicator is illuminated from behind by a replaceable incandescent bulb and bulb holder unit, which is located on the instrument cluster electronic circuit board. The high beam indicator is serviced as a unit with the instrument cluster.

OPERATION

The high beam indicator gives an indication to the vehicle operator whenever the headlamp high beams are illuminated, or when the exterior lamps are inadvertently left On. This indicator is controlled by a hard wired input to the cluster from the headlamp beam select switch circuitry of the left multi-function switch, and by the instrument cluster circuit board based upon cluster programming based upon hard wired inputs from the head/park/fog lamp switch circuitry of the left multi-function switch, from the driver door ajar switch, and from the ignition switch. The high beam indicator bulb is completely controlled by the instrument cluster logic circuit, and that logic will allow this indicator to operate whenever the instrument cluster receives a battery current input on the fused B(+) circuit. Therefore, the bulb can be illuminated regardless of the ignition switch position. The bulb only illuminates when it is provided a path to ground by the instrument cluster transistor. The instrument cluster will turn on the high beam indicator for the following reasons:

- **Beam Select Switch Input** - Each time the cluster detects battery current on the beam select switch sense circuit (beam select switch closed = high beams selected or optical horn feature activated) the high beam indicator will be illuminated solid. This input can occur when the headlamp high beams are selected or when the optical horn feature is activated. The indicator remains illuminated until the beam select switch sense input to the cluster is an open circuit (beam select switch open = high beams not selected and optical horn feature not activated), or until the exterior lighting is turned off, whichever occurs first.
- **Exterior Lamps-On Optical Warning** - Each time the cluster detects battery current on the headlamp switch output circuit (park or head lamp switch closed = exterior lighting is On), ground on the driver door ajar switch sense circuit (driver door ajar switch closed = driver door is open), and the fused ignition switch output (run-start) input is an open circuit (ignition switch is in a position other than On or Start), the high beam indicator will be flashed on and off repeatedly. The indicator will continue to flash until the exterior lighting is turned Off, until the driver door is closed, or until the ignition switch is turned to the On or Start positions, whichever occurs first.
- **Actuator Test** - Each time the cluster is put through the actuator test, the high beam indicator will be turned on, then off again during the bulb check portion of the test to confirm the functionality of the bulb and the cluster control circuitry.

The instrument cluster continually monitors the headlamp beam select switch and circuitry integral to the left multi-function switch, the driver door ajar